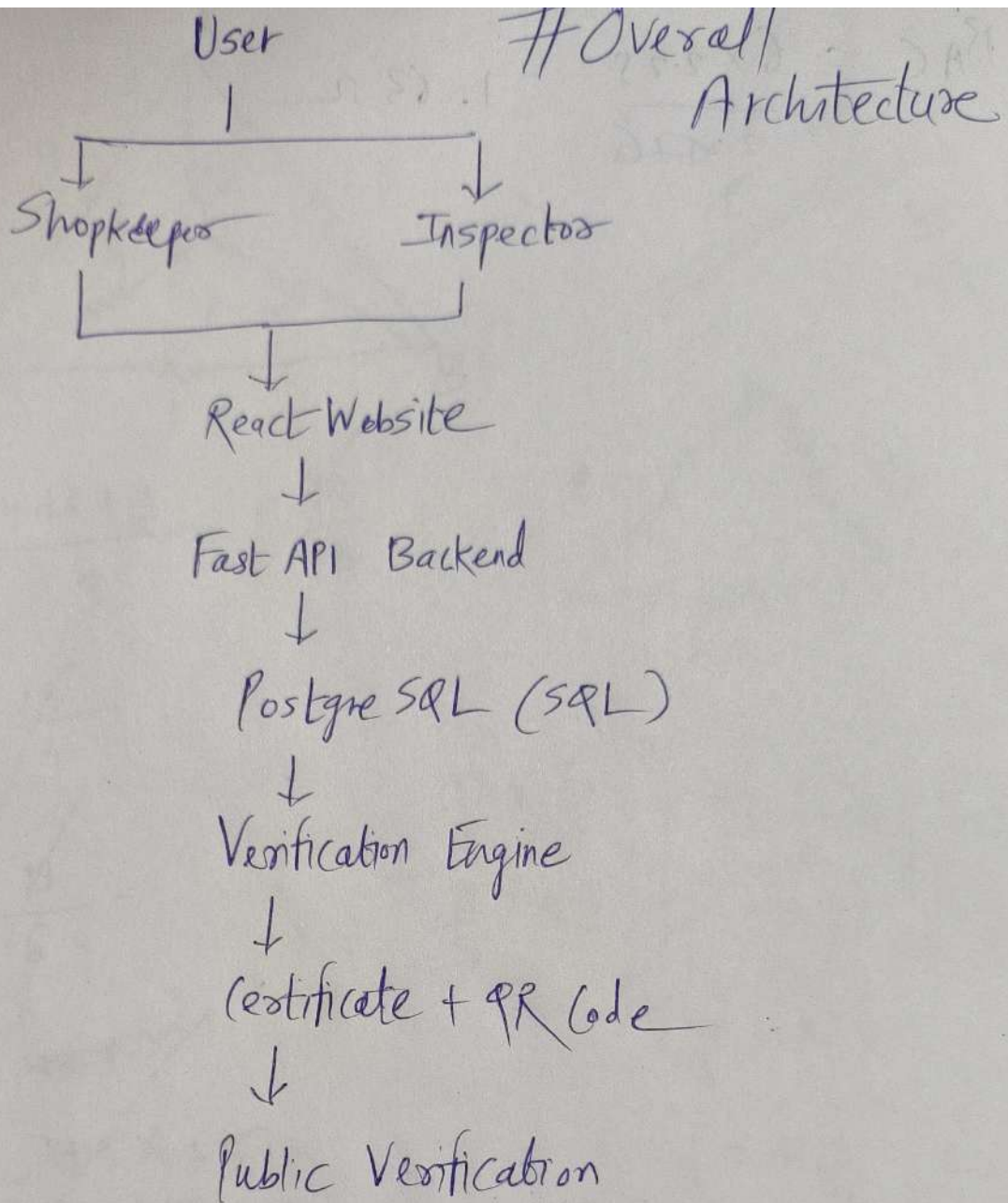




② Owner Side

Dashboard

My Instruments : 4

o Verified - 2

o Pending - 1

o Expired - 1

+ Add instrument



Form :

Instrument Type

Manufacturer

Model

Serial No

Capacity

Location

Purchase date

③ Verification Request

Request Verification

↳ generates System App. Id

Ex: VER-2026-0028

Status :

Inspection Pending

④ Inspector Side

Insp. login :

Pending - 24

Todays Inspection : 8

Approval - 142

Rejected - 7

→ Application opens :

Instrument # Unique Id

↳ Inspector checks physical machine

Enter details :

Stand. Wt : 10 kg

Machine Reading : 10.02 kg

Calculated Error : ~~$\pm 20 \text{ kg}$~~ $\pm 20 \text{ gm}$

↳ There will be an condition that Allowed Error is $\pm 50 \text{ g}$

$\therefore 20 \text{ gm} < 50 \text{ gm}$

Verifig-Pass

⑤ Evidence Upload

(Inspector)
→ Instrument photo
Serial No
Test Setup
Document

∴ Remark for Passed Verification

⑥ Approval

Inspector Approve

Backend Starts Working

Verification status = Verified

Certificate generated = Yes

If Rejects-

Remark will be "Error exceeds permissible limit"

⑦ Digital Certificate

↳ ^{form} To be decided

⑧

QR code

unique qr code generation

Tech Stack

Frontend → React + Tailwind CSS

Backend → Python + FastAPI

Database → PostgreSQL / MySQL / MongoDB

Authentication → Tson Web Token (JWT)

Certificate → Python ReportLAB

QR : Python QR-code library (Numpy, pandas)

~~File Storage / uploads~~